

MDI3003

ADVANCED PREDICTIVE ANALYTICS

Laboratory Assessment – 4

Probabilistic Customer Segmentation and Segment Prediction using Naive Bayes

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Report prepared from the submitted notebook, using the supplied Decision-Tree report as the structural and presentation reference.

Abstract

Customer segmentation is a supervised classification problem in which customer characteristics are used to predict predefined customer segment labels. Accurate segmentation can support the identification of groups with similar demographic and behavioral characteristics and can provide a structured basis for subsequent customer analysis. This study develops a **leakage-safe, reproducible and baseline-first Naive Bayes classification workflow** for predicting four customer segments, namely **A, B, C and D**, using demographic, behavioral and other profile attributes. The workflow is designed to evaluate different Naive Bayes models under consistent preprocessing and validation conditions while also examining model confidence and the reliability of automated predictions.

The dataset used in the notebook contains **8,068 observations and 11 columns**, consisting of a customer identifier, **nine predictor variables**, and the target variable **Segmentation**. The predictors include demographic and profile-related attributes such as Gender, Ever_Married, Age, Graduated, Profession, Family_Size, Work_Experience, Spending_Score and Var_1. The customer identifier is explicitly excluded from the predictor matrix to prevent it from influencing the classification process. The dataset contains missing values, particularly in **Work_Experience, Family_Size, Ever_Married, Profession, Graduated and Var_1**. The notebook applies preprocessing within the modelling workflow to reduce the possibility of information leakage. It also verifies the data structure by checking for duplicate customer IDs and exact duplicate records, with no duplicate IDs or exact duplicate rows identified.

The target variable consists of four customer segments, **A, B, C and D**, with a moderate degree of class imbalance and Segment D representing the largest class. Because class imbalance can make overall accuracy insufficient as a single evaluation measure, the study emphasizes **Macro F1**, together with accuracy and weighted F1. Macro F1 gives equal importance to each segment and is therefore particularly useful for evaluating whether the classifier performs consistently across all four customer groups.

Four classification approaches were evaluated: **DummyClassifier, GaussianNB, BernoulliNB and CategoricalNB**. The DummyClassifier was included as a baseline to establish the performance that could be obtained without learning meaningful relationships from the predictors. GaussianNB was evaluated using numerical representations, while BernoulliNB and CategoricalNB were evaluated using transformed categorical and discretized representations. The preprocessing strategy included imputation of missing values, encoding of categorical variables and discretization of selected numerical variables. During experimentation, the CategoricalNB pipeline was corrected to ensure that all transformed feature values were **non-negative**, satisfying the model's input requirements.

Model performance was assessed using **stratified five-fold cross-validation**, allowing the models to be compared under consistent train-validation splits. Among the evaluated models, **CategoricalNB achieved the strongest cross-validation performance**, obtaining a mean Macro F1 of **0.4845** with a standard deviation of **0.0101**. The combined feature group containing the available predictors also produced the best feature-group performance, with a

Macro F1 of **0.4845**. The results indicate that incorporating the complete transformed feature representation provided more useful information than relying on narrower feature groups alone. Although the performance demonstrates useful predictive information beyond the baseline, the moderate Macro F1 also indicates that the classification problem remains challenging, particularly for individual segments.

Following cross-validation, CategoricalNB was selected for evaluation on a **locked test set** that was kept separate from model selection. On this test set, the selected model achieved an **accuracy of 51.18%**, a **Macro F1 of 0.4853**, and a **weighted F1 of 0.4955**. Per-class analysis showed differences in predictive performance between the four segments. Segment D obtained the strongest recall and F1 performance, whereas Segment B showed substantially lower recall. This indicates that the overall accuracy should not be interpreted as uniform performance across all customer segments and reinforces the importance of class-wise evaluation.

In addition to conventional classification metrics, the notebook performs a **confidence analysis using out-of-fold predicted probabilities**. The analysis examines whether predictions associated with higher confidence are more likely to be correct. The results show a general tendency for observed accuracy to increase as prediction confidence increases, although the relationship is not perfectly monotonic across every confidence interval. This finding provides a basis for introducing a review or abstention mechanism rather than requiring every prediction to be accepted automatically.

A validation-derived **frozen review threshold of 0.55** was subsequently established. Predictions with confidence below this threshold are sent for manual review, while predictions meeting or exceeding the threshold are accepted automatically. At this threshold, the validation analysis achieved approximately **55.38% automatic coverage with 61.89% accuracy among accepted cases**. When the same frozen threshold was applied to the locked test set, **910 out of 1,614 customers** were automatically predicted, corresponding to **56.38% automatic coverage**, while **704 customers (43.62%)** were sent for review. This demonstrates the trade-off between prediction coverage and reliability: accepting fewer predictions allows the system to concentrate automated decisions on cases for which the model expresses greater confidence.

Overall, the study demonstrates a complete customer segmentation classification workflow covering **data inspection, leakage prevention, preprocessing, baseline comparison, model compatibility, stratified cross-validation, locked-test evaluation, per-class analysis, confidence calibration analysis and prediction review**. The results identify CategoricalNB with the combined feature representation as the strongest evaluated Naive Bayes configuration. However, the moderate predictive performance and the existence of low-confidence and misclassified cases indicate that the model should be used as an **analytical segmentation aid rather than an unquestionable automated decision mechanism**. The proposed confidence-based review policy provides an additional safeguard by allowing uncertain predictions to be examined rather than forcing automatic classification in every case.

Keywords: Customer Segmentation, Supervised Classification, Naive Bayes, CategoricalNB, BernoulliNB, GaussianNB, Macro F1, Weighted F1, Stratified Cross-Validation, Locked Test Set, Confidence Analysis, Prediction Uncertainty, Abstention, Human Review, Leakage Prevention.

Introduction

Predictive customer segmentation is an important application of supervised machine learning in which information available about individual customers is used to estimate a predefined customer segment. Such predictions can support customer profiling, marketing analysis, service personalization and the identification of groups with similar characteristics. In a predictive segmentation setting, the objective is not to discover previously unknown groups, but to learn patterns from customers whose segment labels are already known and use those patterns to classify new observations.

The present study treats customer segmentation as a **supervised multiclass classification problem**. This distinction is important because customer segmentation can also be approached through unsupervised clustering techniques. In clustering, the algorithm discovers groups without a predefined target variable. In contrast, the dataset used in this notebook contains an explicit **Segmentation** target with four known classes: **A, B, C and D**. Therefore, the primary objective is to learn a mapping between the available customer attributes and these predefined segment labels.

The analysis follows a structured and reproducible machine-learning workflow. The workflow begins with dataset inspection and data-quality auditing, followed by identification of missing values, duplicate records, identifier-related issues and possible sources of information leakage. Since customer ID is an identifier rather than a meaningful predictive attribute, it is retained for traceability and validation checks but excluded from the model's predictor matrix. The target variable is similarly kept separate from the input features.

A major emphasis of the notebook is **leakage prevention**. Information from the target or from observations that should remain unseen during evaluation must not influence the training process. To address this, the analysis separates the available data into training and locked-test observations and applies preprocessing in a model-compatible manner. The locked test set is reserved for final evaluation rather than being repeatedly used for model selection. This design provides a more reliable estimate of how the selected classifier behaves on previously unseen customer records.

The modelling stage follows a **baseline-first strategy**. A DummyClassifier establishes a simple reference level against which more meaningful machine-learning models can be compared. Three Naive Bayes approaches—**GaussianNB**, **BernoulliNB** and **CategoricalNB**—are then evaluated using representations appropriate to their underlying assumptions. The notebook uses model-specific preprocessing because the different Naive Bayes algorithms have different requirements regarding the form and range of input features.

Model comparison is performed using **five-fold stratified cross-validation**. Stratification ensures that the four customer segments are represented consistently across the

validation folds. The primary model-selection criterion is **mean cross-validation Macro F1**, rather than accuracy alone. This is particularly relevant because the target distribution is not perfectly balanced. Macro F1 calculates performance across the individual classes and gives each segment equal importance, helping to prevent strong performance on the largest class from masking weak performance on smaller classes.

The study also evaluates different **feature groups**, including demographic, behavioral and combined representations. This allows the analysis to determine whether predictive performance depends primarily on a particular category of customer attributes or whether combining information from multiple groups provides a stronger representation. All feature-group comparisons follow the same validation framework to maintain consistency between experiments.

After model selection, the chosen classifier is evaluated on the **locked test set**. Both aggregate and class-wise metrics are considered. Aggregate measures provide an overall view of predictive performance, while per-class precision, recall and F1 scores reveal whether the model performs consistently for segments A, B, C and D. This distinction is important because a model may achieve reasonable overall accuracy while still performing poorly for a particular customer segment.

The notebook extends the evaluation beyond conventional classification metrics by examining **prediction confidence and the coverage-error trade-off**. Instead of automatically accepting every prediction, the analysis investigates whether low-confidence predictions should be separated for further review. Out-of-fold probabilities are used to study the relationship between model confidence and observed prediction accuracy. A validation-derived confidence threshold is then used to establish an abstention or review mechanism. Under the selected threshold of **0.55**, predictions below the threshold can be referred for review while sufficiently confident predictions can be accepted automatically.

This approach is particularly useful because a classification model does not necessarily provide equally reliable predictions for every customer. Some observations may have a clear probability advantage for one segment, while others may have probabilities that are distributed relatively evenly across multiple segments. Treating both situations identically could result in unnecessary automated errors. The review mechanism therefore provides a way of balancing **automation coverage and prediction reliability**.

The final stage includes representative error analysis and a demonstration of prediction for a new customer profile. These analyses provide an interpretable view of how the model behaves beyond aggregate numerical scores. They also help identify situations in which model predictions should be treated cautiously.

The reference report supplied for this work follows an **industry-style laboratory structure**, covering problem framing, dataset description, data auditing, leakage prevention, train/test design, preprocessing, model comparison, evaluation, interpretation, responsible use, reproducibility and conclusion. The present report follows the same overall organization and level of detail while replacing the original medical Decision Tree

context with the actual customer-segmentation problem and the Naive Bayes models implemented in the notebook.

Overall, the introduction establishes customer segmentation as a controlled supervised-learning problem and provides the foundation for evaluating whether Naive Bayes models can provide useful predictive information from customer profile attributes while maintaining appropriate validation, leakage prevention, confidence analysis and responsible-use practices.

Objectives

The major objectives of this study are:

- **Formulate customer segmentation as a supervised multiclass classification problem** in which the predefined target variable, Segmentation, contains four classes: A, B, C and D.
- **Audit the dataset systematically** for missing values, duplicate records, duplicate customer identifiers, feature cardinality and potential label circularity before model development.
- **Remove direct customer identifiers from the predictor matrix** so that customer IDs are used only for traceability and data-integrity checks rather than as predictive variables.
- **Maintain a locked test set** that remains separate from model-selection decisions and is used only for final evaluation of the selected classifier.
- **Construct model-specific preprocessing pipelines** that produce representations compatible with GaussianNB, BernoulliNB and CategoricalNB while handling missing values and categorical/numerical variables appropriately.
- **Establish a DummyClassifier baseline** and compare it against the three Naive Bayes classifiers under a common evaluation framework.
- **Compare the models using identical five-fold stratified cross-validation**, ensuring that model comparisons are based on consistent validation conditions.
- **Select the final model using mean cross-validation Macro F1**, thereby giving equal importance to the predictive performance of all four customer segments rather than relying only on overall accuracy.
- **Compare demographic, behavioral and combined feature groups** using the same validation folds to determine the contribution of different types of customer attributes.
- **Evaluate the selected model on unseen locked-test observations** using accuracy, Macro F1, weighted F1 and class-wise precision, recall and F1 measures.
- **Analyze prediction confidence** using out-of-fold probabilities to determine whether higher-confidence predictions generally correspond to higher observed accuracy.
- **Study the coverage-error trade-off** and determine how restricting automatic predictions affects the number of accepted cases and their observed accuracy.
- **Establish a validation-derived review threshold** that allows low-confidence predictions to be referred for review instead of being automatically accepted.

- **Analyze representative classification errors** to understand the circumstances associated with incorrect or uncertain predictions.
- **Demonstrate prediction for new customer profiles** while clearly reporting the associated confidence and review status.
- **Document limitations and responsible-use considerations**, emphasizing that the classifier is intended as an analytical customer-segmentation aid rather than an unquestionable automated business decision-maker.

Problem Statement and Prediction Context

The central problem addressed in this study is the prediction of a customer's predefined segment using routinely available customer profile attributes. Each observation represents a customer record containing a collection of demographic, profile and behavioral characteristics. The target variable, **Segmentation**, assigns each customer to one of four predefined classes: **A, B, C or D**.

The task can therefore be expressed as a multiclass classification problem: where:

- represents the customer's observed attributes,
- represents the predefined customer segment,
- .

The objective of the classifier is to learn the relationship between the predictor variables and the known segment labels from the training data and then use the learned relationship to predict the segment of previously unseen customers.

The final predictor matrix contains **nine variables**:

1. **Gender**
2. **Ever_Married**
3. **Age**
4. **Graduated**
5. **Profession**
6. **Family_Size**
7. **Work_Experience**
8. **Spending_Score**
9. **Var_1**

The **Customer ID** is deliberately excluded from the predictor matrix. Although the identifier is useful for identifying individual records, checking train/test overlap and tracing predictions back to customers, it does not represent a customer characteristic that should determine the predicted segment. Including it as a feature could allow the model to learn meaningless identifier-specific patterns and would reduce the interpretability of the resulting classifier. Similarly, the **Segmentation** variable is treated exclusively as the target and is not included among the predictors. This separation ensures that the model attempts to predict the segment from customer characteristics rather than receiving the answer directly during training.

The prediction context is therefore:

The notebook further recognizes that not every prediction has the same level of reliability. A customer may receive a strong probability for one segment, while another customer may have

relatively similar probabilities across several segments. Consequently, the study does not evaluate the classifier solely according to whether it produces a prediction for every observation. It additionally considers **prediction confidence, automatic coverage and review requirements**.

The final intended use is therefore **analytical customer-segmentation prediction**. The model can be used to support exploratory analysis and provide an estimated segment for customer records, but its predictions should not be interpreted as definitive ground truth. In particular, low-confidence cases can be separated for review using the proposed confidence-based mechanism.

This framing provides the basis for the subsequent sections of the report, which examine the dataset, data quality, preprocessing, model comparison, validation performance, locked-test performance, confidence behavior, error patterns and responsible use of the resulting segmentation model.

Dataset Description

The dataset used in this study represents customer profiles for supervised customer-segmentation prediction. Each observation corresponds to an individual customer, identified by a unique customer ID, together with demographic, behavioral and other profile attributes. The prediction target is the predefined **Segmentation** variable, which contains four classes: **A, B, C and D**.

The dataset contains **8,068 observations and 11 columns**. Of the 11 columns, one is the customer identifier (ID), one is the target (Segmentation), and the remaining **nine columns are used as predictors**. The customer ID is retained for traceability and data-integrity checks but is deliberately excluded from the machine-learning feature matrix.

The dataset audit confirms that all **8,068 customer IDs are unique**, with no duplicate IDs. Similarly, no exact duplicate rows were identified. These checks reduce concerns that repeated observations could artificially influence the training or evaluation results.

Attribute	Value
Observations	8,068
Columns	11
Predictors	9
Identifier	ID
Target	Segmentation
Target classes	A, B, C, D
Exact duplicate rows	0
Unique IDs	8,068
Duplicate IDs	0

Table 1. Dataset summary extracted from the submitted notebook.

Feature taxonomy

Group	Features
Demographic	Gender, Ever_Married, Age, Graduated, Profession, Family_Size
Behavioral	Work_Experience, Spending_Score
Other	Var_1
Psychographic	No explicitly documented variables

The nine predictor variables are organized into demographic, behavioral and other feature groups. This grouping is also used later in the notebook to compare the predictive contribution of different feature categories.

Demographic Features

The demographic/profile group consists of:

Gender
Ever_Married
Age
Graduated
Profession
Family_Size

These variables describe relatively stable or profile-oriented characteristics of customers. They provide information about age, gender, marital status, educational status, profession and family structure.

Behavioral Features

The behavioral group contains:

Work_Experience
Spending_Score

These variables provide information related to the customer's work background and spending behavior. In particular, Spending_Score provides a behavioral characteristic that may help distinguish between customer segments.

Other Feature

The remaining predictor is:

Var_1

The notebook does not provide a clearly documented semantic interpretation for this variable. It is therefore retained as an available predictor, while its provenance is treated cautiously in the leakage and circularity audit.

Psychographic Features

No explicitly documented psychographic-only variables are present in the dataset. Therefore, the notebook does not perform a separate psychographic feature-group ablation experiment.

The grouping provides a structured way to evaluate whether customer segmentation can be predicted effectively from a particular category of information or whether combining different types of customer attributes provides greater predictive value.

Target distribution

The prediction target is the Segmentation variable, which contains four classes: The dataset is moderately imbalanced. **Segment D is the largest class**, whereas Segments A, B and C have broadly comparable proportions. This imbalance is an important consideration for both model training and evaluation.

A model that predicts the largest class particularly well could obtain a reasonable overall accuracy while performing poorly on one or more smaller classes. To avoid this issue, the study emphasizes **Macro F1** during model selection. Macro F1 calculates the F1 score independently for each segment and then gives equal importance to all four classes.

The target distribution is also preserved during the train/test split through stratified sampling. This ensures that the training and locked-test partitions maintain approximately the same class proportions as the complete dataset.

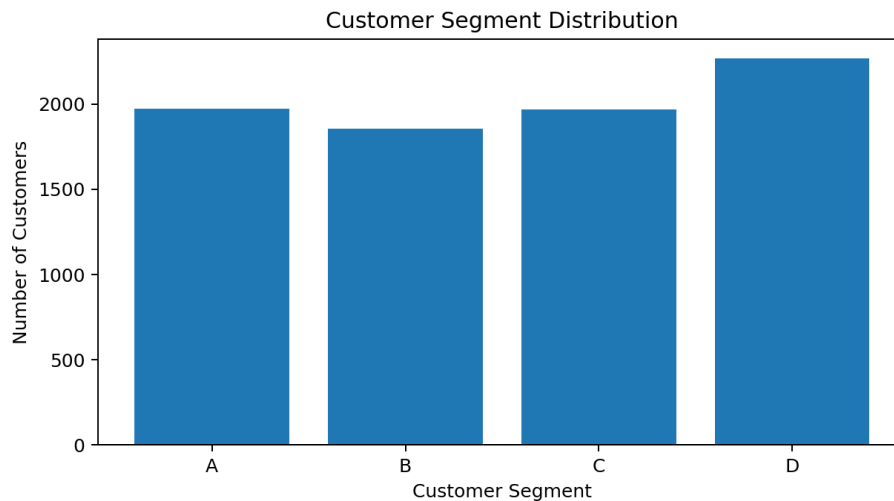


Figure 1. Distribution of the four customer segments. Segment D is the largest class, while A, B and C have broadly similar proportions.

Data Audit and Exploratory Analysis

Before model construction, the dataset was subjected to a structured data-quality audit. The purpose of this stage was to identify missing values, duplicate records, identifier problems, numerical-range issues and possible sources of information leakage.

The audit confirmed that there are **no exact duplicate rows** and that all **8,068 customer IDs are unique**. Therefore, there is no evidence of duplicated customer identifiers or repeated complete observations in the submitted dataset.

The missing-value analysis identified missing observations in six predictors. The largest amount of missingness occurs in *Work_Experience*, followed by *Family_Size*. Smaller amounts of missingness occur in *Ever_Married*, *Profession*, *Graduated* and *Var_1*.

Table 2. Missingness summary

Feature	Missing	Percent
Work_Experience	829	10.28%
Family_Size	335	4.15%
Ever_Married	140	1.74%
Profession	124	1.54%
Graduated	78	0.97%
Var_1	76	0.94%
Gender	0	0.00%
Age	0	0.00%
Spending_Score	0	0.00%

Work_Experience has the highest missingness, with **829 missing observations corresponding to 10.28% of the dataset**. Family_Size is the next most affected feature, with 335 missing observations or 4.15%.

The remaining affected variables have less than 2% missingness individually. In contrast, Gender, Age and Spending_Score contain no missing values.

The missing values are not removed through complete-case deletion. Instead, missing-value handling is incorporated into the preprocessing pipelines so that observations can be retained for model development. Numerical and categorical variables are handled separately according to their data types.

Figure 2. Missing-value percentage by predictor.

The missingness pattern indicates that Work_Experience deserves particular attention because it contributes substantially more missing information than the other variables. However, because the notebook incorporates imputation into the preprocessing workflow, the feature remains available for model training.

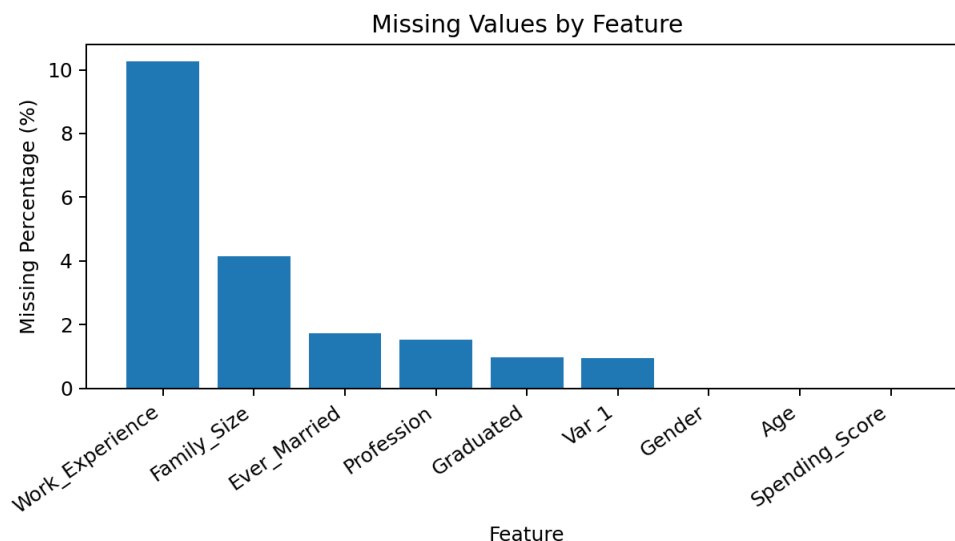


Figure 2. Missing-value percentage by predictor.

Numerical range checks

Feature	Mean	Std. Dev.	Min	Max
Age	43.47	16.71	18	89
Work_Experience	2.64	3.41	0	14
Family_Size	2.85	1.53	1	9

The ranges observed in the notebook do not show negative values for the three numerical predictors. Work_Experience has the highest missingness and is therefore handled inside the model pipeline.

Leakage and circularity audit

Information leakage is a major concern in supervised machine learning because it can produce artificially high validation performance. Leakage occurs when information that would not legitimately be available when predicting a new customer is indirectly provided to the model.

The notebook therefore separates the customer identifier and target variable from the predictor matrix. ID is used only for traceability and train/test overlap checks, while Segmentation is used exclusively as the prediction target.

The nine predictors were also reviewed for potential direct label-proxy relationships. The notebook records **no documented evidence of direct label leakage** for:

- Gender
- Ever_Married
- Age
- Graduated
- Profession
- Family_Size
- Work_Experience
- Var_1

However, the notebook explicitly identifies **Spending_Score as requiring label-provenance confirmation**. Var_1 is also retained pending provenance confirmation. These points should therefore be considered limitations of the current analysis rather than treated as fully resolved leakage questions.

The provenance issue is especially important if the model is intended for practical deployment. A feature can appear predictive without being a legitimate predictor if it was generated using information related to the target. Therefore, the present results should be interpreted within the provenance assumptions documented in the notebook.

Train/Test Split and Validation Strategy

The notebook creates an 80:20 stratified split with random seed 42. The training partition contains 6,454 observations and the locked test partition contains 1,614 observations. Class proportions remain almost identical across the two partitions. Customer IDs were checked before modeling and there was zero overlap between training and test IDs.

Parameter	Value
Total usable observations	8,068
Training observations	6,454 (79.995%)
Locked test observations	1,614 (20.005%)
Sampling	Stratified
Cross-validation	5-fold StratifiedKfold
Random state	42
Test-set use during selection	No

Table 3. Train/test and validation configuration.

The stratified split ensures that the four target classes remain represented in approximately the same proportions in both partitions. This is particularly important because the dataset has moderate class imbalance.

The notebook also performs a customer-ID overlap check between the two partitions. The result shows zero overlap, confirming that the same customer identifier does not occur in both training and locked-test sets.

The training data is subsequently used for five-fold stratified cross-validation. In this process, the training set is divided into five folds while preserving the class distribution. Each fold is used once as validation data while the remaining folds are used for training.

The locked test set remains untouched during model selection. This distinction is important because repeatedly evaluating candidate models on the test set would effectively turn the test set into another validation set and could lead to optimistic performance estimates.

Leakage-Safe Preprocessing

Preprocessing is implemented within **scikit-learn pipelines**, allowing transformations to be fitted as part of the model-training workflow rather than being calculated using the complete dataset before validation.

The nine predictors are divided into numerical and categorical variables.

Numerical Features

The numerical features are:

- Age
- Family_Size
- Work_Experience

These variables require numerical preprocessing and missing-value treatment.

Categorical Features

The categorical features are:

- Gender
- Ever_Married
- Graduated
- Profession
- Spending_Score
- Var_1

The notebook applies **median imputation for numerical variables** and **most-frequent imputation for categorical variables**. This ensures that missing observations can be retained while providing each model with a complete transformed representation.

Importantly, preprocessing differs according to the assumptions of the selected Naive Bayes classifier.

GaussianNB Representation

GaussianNB requires numerical input and assumes that features within each class can be represented using Gaussian distributions. Consequently, the categorical variables must be transformed into numerical representations before they can be used by the classifier.

BernoulliNB Representation

For BernoulliNB, numerical variables are **quantile-binned**, while categorical variables are **one-hot encoded**. The resulting representation is compatible with the Bernoulli Naive Bayes formulation. Quantile binning converts numerical observations into discrete ranges, while one-hot encoding represents categorical levels using binary indicator variables.

CategoricalNB Representation

CategoricalNB requires categorical feature representations. Therefore, the numerical variables are **quantile-binned into ordinal categories**, while the categorical variables are **ordinal encoded**.

An important preprocessing issue was identified during the initial CategoricalNB experiment. The transformed representation initially did not satisfy the requirement that categorical values be non-negative. As a result, the first cross-validation attempt produced invalid/NaN metrics.

The preprocessing pipeline was subsequently corrected by shifting the encoded representation so that all transformed values were non-negative. The notebook explicitly verifies that the minimum transformed value is **0.0**, confirming that the corrected representation satisfies the model requirement.

After the correction, one validation fold was successfully tested and produced a **Macro F1 of 0.496391**. The complete five-fold cross-validation was then rerun, resulting in:

- **Mean Macro F1 = 0.484523**
- **Standard deviation = 0.010115**

This correction is significant because it demonstrates that model comparison was not based on the invalid initial representation. Instead, the preprocessing was checked, corrected and subsequently re-evaluated before the final CategoricalNB result was used.

Importance of Model-Specific Preprocessing

The preprocessing strategy is particularly important in this study because the three Naive Bayes models do not operate under identical feature assumptions.

The same raw customer attributes can therefore produce different transformed representations depending on the classifier being evaluated. This means that model performance reflects not only the underlying classifier but also the suitability of the feature representation for that classifier.

The notebook consequently avoids forcing all three models to use an identical transformed matrix. Instead, each model receives a representation compatible with its assumptions:

This approach provides a more appropriate comparison of the Naive Bayes alternatives while maintaining the same underlying customer information.

Model Development and Cross-Validation

The model-development stage evaluates whether customer attributes contain sufficient predictive information to distinguish the four predefined customer segments. Since the target variable contains four classes—A, B, C and D—the problem is treated as a **multiclass classification problem**.

Four models were evaluated:

1. **DummyClassifier**
2. **Gaussian Naive Bayes (GaussianNB)**

3. Bernoulli Naive Bayes (BernoulliNB)

4. Categorical Naive Bayes (CategoricalNB)

All models were evaluated using the **same five stratified cross-validation folds**. Using identical folds makes the comparison more consistent because each model is evaluated on the same validation partitions.

The DummyClassifier provides a baseline by always predicting the most frequent class. It does not learn meaningful relationships between customer attributes and the segmentation target. GaussianNB uses numerical representations, BernoulliNB operates on binary/binning representations, and CategoricalNB is designed for categorical representations and was implemented using the corrected mixed-feature preprocessing pipeline.

Cross-Validation Metrics

The primary selection criterion is **mean Macro F1**. Macro F1 is particularly suitable because the target classes are not perfectly balanced. It calculates the F1 score separately for each class and then averages the four class-level scores, giving each segment equal importance.

Model	Accuracy	Macro F1	Weighted F1	Macro F1 SD
CategoricalNB	0.5094	0.4845	0.4938	0.0101
BernoulliNB	0.5031	0.4801	0.4891	0.0087
GaussianNB	0.4202	0.3719	0.3829	0.0119
Dummy	0.2811	0.1097	0.1233	0.0001

Table 4. Corrected five-fold cross-validation comparison.

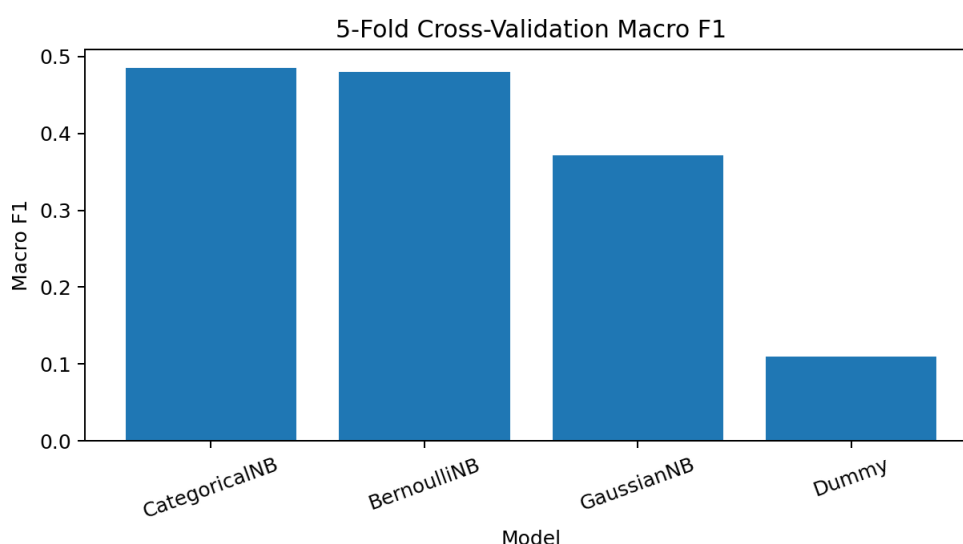


Figure 3. Mean cross-validation Macro F1. CategoricalNB ranks first, followed closely by BernoulliNB.

The results show that **CategoricalNB achieved the highest Macro F1 of 0.4845**, followed by BernoulliNB with 0.4801. GaussianNB achieved a considerably lower Macro F1 of 0.3719, while the DummyClassifier produced only 0.1097.

The DummyClassifier provides an important reference point. Its accuracy of 0.2811 reflects the effect of predicting the dominant class rather than learning meaningful patterns. The substantially higher performance of all three Naive Bayes models indicates that the customer attributes contain predictive information beyond the majority-class baseline.

CategoricalNB achieved an accuracy of 0.5094 and weighted F1 of 0.4938. Its Macro F1 standard deviation was 0.0101, indicating relatively stable performance across the five validation folds.

BernoulliNB performed very closely to CategoricalNB. Its Macro F1 of 0.4801 is only approximately **0.0045 lower** than the CategoricalNB result. Therefore, although CategoricalNB ranks first, the difference is relatively small and should not be interpreted as evidence of decisive practical superiority.

GaussianNB performed substantially worse. Its Macro F1 of 0.3719 is approximately 0.1126 below CategoricalNB. This suggests that the Gaussian representation was less suitable for the mixed demographic and categorical structure of the customer data.

Interpretation of Model Ranking

The resulting ranking according to Macro F1 is:

The selection of CategoricalNB is therefore based on the predefined model-selection criterion rather than simply selecting the model with the highest accuracy.

Figure 3. Mean cross-validation Macro F1. CategoricalNB ranks first, followed closely by BernoulliNB, while GaussianNB and the DummyClassifier perform considerably lower.

Feature-Group Experiment

After comparing different classifiers, the notebook investigates whether different groups of customer attributes contribute differently to segmentation prediction.

Three feature configurations were evaluated using the selected **CategoricalNB** classifier:

- **Demographic features**
- **Behavioral features**
- **Combined features**

The same cross-validation folds were maintained for all feature-group experiments. This ensures that differences in performance are associated with the feature groups rather than differences in validation samples.

Feature Group	Features	Accuracy	Macro F1	SD
Combined	9	0.5094	0.4845	0.0101

Demographic	6	0.5033	0.4802	0.0085
Behavioral	2	0.4070	0.2726	0.0220

Table 5. Feature-group comparison.

The **combined feature group** achieves the best performance, with a Macro F1 of 0.4845. This indicates that using all nine predictors provides the strongest overall predictive representation among the tested feature groups.

The demographic-only group performs almost as well, achieving a Macro F1 of 0.4802. The difference between demographic and combined features is relatively small:

$$0.4845 - 0.4802 = 0.0043$$

This indicates that demographic attributes account for a substantial portion of the predictive information available to the model.

In contrast, the behavioral-only group performs considerably worse. Its Macro F1 is only **0.2726**, compared with 0.4845 for the combined representation. Its accuracy is also lower at 0.4070.

Therefore, the notebook results suggest that behavioral features alone are insufficient for accurately separating the four customer segments. However, they can still contribute additional information when combined with demographic and other attributes.

It is important to interpret this as a predictive association rather than a causal conclusion. The experiment demonstrates how the model performs with different feature groups; it does not establish that demographic characteristics cause customer segmentation.

Figure 4. Macro F1 by feature group. The combined feature representation performs best, followed closely by the demographic group, while behavioral features alone show substantially weaker predictive performance.

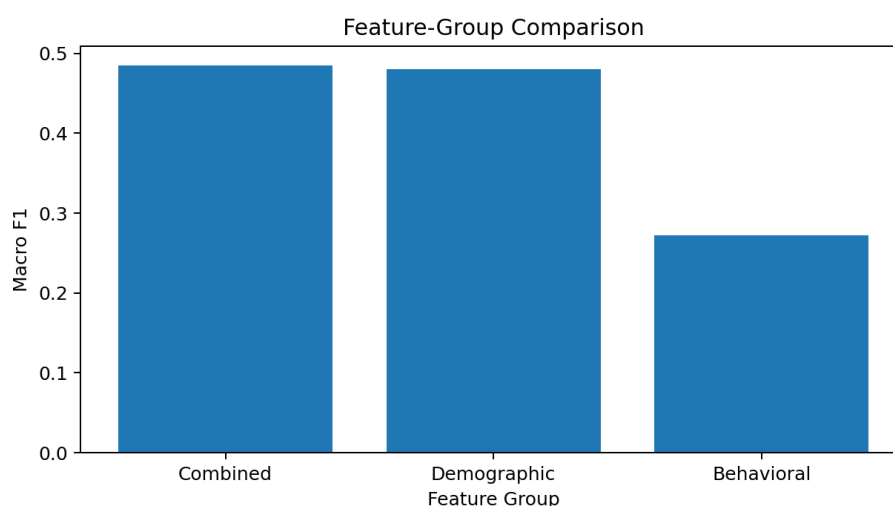


Figure 4. Macro F1 by feature group.

Final Model Selection and Locked-Test Evaluation

Based on the five-fold cross-validation experiment, the final model is:

CategoricalNB + Combined feature group

The model uses all nine available predictors and is selected according to the highest mean five-fold validation Macro F1.

Importantly, the locked test set was not used during this selection process. This protects the final evaluation from test-set-driven model selection

Table 6. Frozen model configuration

Item	Selected configuration
Classifier	CategoricalNB
Feature group	Combined
Number of predictors	9
Selection metric	Mean 5-fold CV Macro F1
CV Macro F1	0.484523
CV Macro F1 SD	0.010115
CV Weighted F1	0.493837
Test used for selection	False

The final model configuration is considered **frozen** after selection. No additional model-selection decisions are made using the locked test observations.

This separation creates two distinct stages:

Model-development stage

The training data is used for:

- preprocessing,
- cross-validation,
- model comparison,
- feature-group comparison,
- model selection,
- confidence-threshold selection.

Final evaluation stage

The locked test set is used only after the model configuration has been frozen.

This provides a more reliable estimate of how the selected model performs on previously unseen observations.

Final locked-test results

After the model configuration was frozen, the pipeline was fit on the complete training partition and evaluated once on the locked test set. The final test accuracy is 0.511772, Macro F1 is 0.485284 and weighted F1 is 0.495491. The closeness of CV and test Macro F1 (0.484523 versus 0.485284) indicates that the selected model's aggregate performance is broadly consistent across the validation and held-out test partitions.

Metric	CV	Locked Test
Accuracy	0.509450	0.511772
Macro F1	0.484523	0.485284
Weighted F1	0.493837	0.495491

Table 7. Cross-validation versus locked-test performance

Class-Wise Evaluation and Confusion Matrix

The class-wise report shows substantial differences among the four segments. Segment D has the strongest recall (0.7225) and F1-score (0.6840), while Segment B has the weakest recall (0.2231) and F1-score (0.2785). Segment C has recall of 0.6548 and F1-score of 0.5670. This indicates that the model is considerably more effective for some segments than others despite the overall Macro F1 of 0.4853.

Class	Precision	Recall	F1	Support
A	0.4255	0.3985	0.4115	394
B	0.3705	0.2231	0.2785	372
C	0.5000	0.6548	0.5670	394
D	0.6495	0.7225	0.6840	454

Table 8. Locked-test class-wise performance.

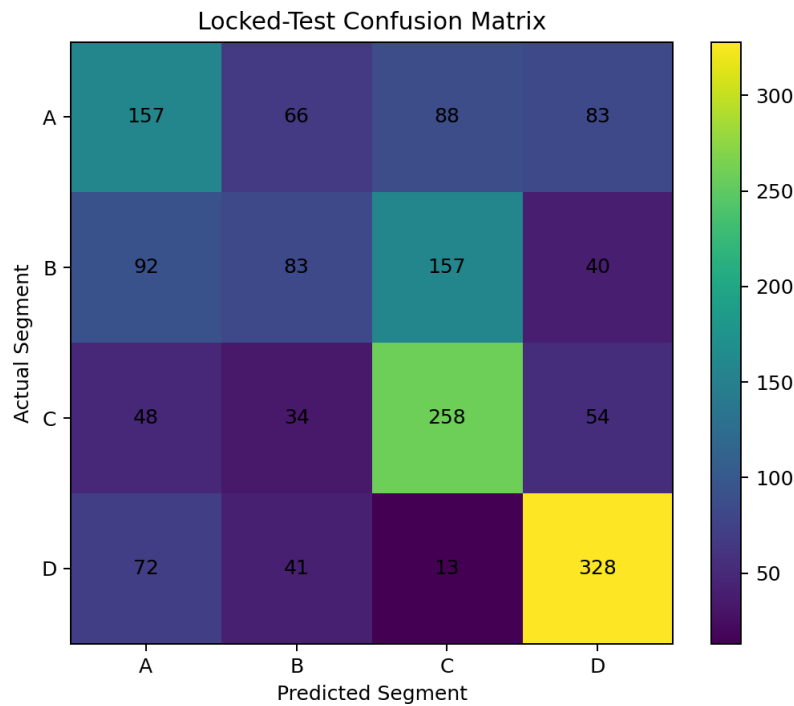


Figure 5. Confusion matrix for the 1,614 locked-test customers.

The confusion matrix contains 826 correct and 788 incorrect predictions. The largest error pattern is B → C (157 cases), followed by B → A (92), A → C (88), A → D (83) and D → A (72). These patterns show that the model has difficulty separating several neighboring segment profiles, particularly Segment B.

Error Analysis and Confidence Analysis

The notebook constructs an individual test-prediction table containing actual segment, predicted segment, prediction confidence, second-best probability and probability margin. Five representative low-confidence errors were examined. Their confidence values are approximately 0.321, 0.323, 0.324, 0.334 and 0.326, with very small probability margins. These cases illustrate that several mistakes occur when the model assigns similar posterior probabilities to competing segments.

Representative error patterns

Error Pattern	Count	Interpretation
B → C	157	Segment B is frequently confused with C.
B → A	92	Segment B is also confused with A.
A → C	88	Some A profiles resemble C.
A → D	83	Some A profiles resemble D.
D → A	72	Some D profiles resemble A.

Out-of-fold confidence

Out-of-fold probabilities were generated on the training data so that confidence analysis did not rely on the locked test set. The OOF accuracy was 0.509452. Observed accuracy generally increased at higher confidence levels: the 0.25–0.40 bin had 29.70% accuracy, while the 0.90–1.00 bin had 72.09% accuracy. The relationship is not perfectly monotonic, which cautions against interpreting posterior confidence as certainty.

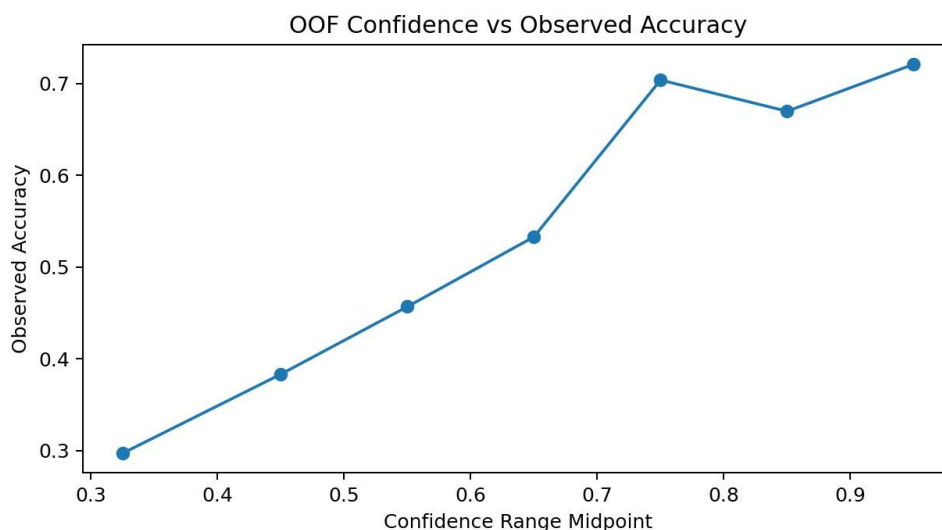


Figure 6. Observed accuracy across out-of-fold confidence ranges.

Review / Abstention Policy

The notebook converts confidence into an operational review policy. Thresholds from 0.20 to 0.90 were evaluated using training out-of-fold predictions. The selected threshold is 0.55, defined by the rule 'lowest threshold achieving at least 60% validation accuracy'. At threshold 0.55, validation coverage is 55.38%, accuracy among accepted cases is 61.89%, and the error rate is 38.11%.

Threshold	Validation Coverage	Validation Accuracy	Error Rate
0.40	90.61%	53.15%	46.85%
0.50	65.54%	58.82%	41.18%
0.55	55.38%	61.89%	38.11%
0.60	45.09%	64.78%	35.22%
0.70	30.73%	70.15%	29.85%
0.90	16.66%	72.09%	27.91%

Table 9. Selected points from the validation coverage–error analysis.

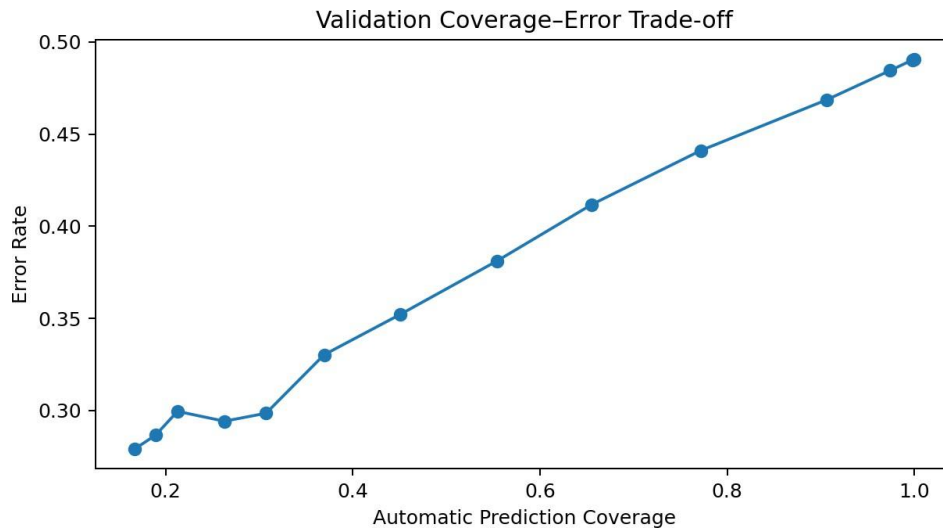


Figure 7. Coverage decreases as the confidence threshold increases, while accepted-case error generally decreases.

Locked-test operational effect

Applying the frozen threshold of 0.55 to the locked test set produces 910 automatic predictions and 704 review cases. Automatic coverage is 56.38%, while the review rate is 43.62%. The review rule therefore avoids forcing an automatic segment assignment for a substantial portion of uncertain cases.

New-Customer Prediction

The notebook demonstrates the frozen pipeline on a new customer profile. The predicted segment is B with confidence 0.493992, below the frozen review threshold of 0.55. Therefore, the operational decision is REVIEW rather than automatic acceptance.

Output	Value
Predicted segment	B
Confidence	0.493992
Review threshold	0.55
Operational decision	REVIEW
P(A)	0.373492
P(B)	0.493992
P(C)	0.117672
P(D)	0.014843

Table 10. Notebook new-customer prediction output.

The prediction probabilities show that B is only moderately ahead of A, which supports the review decision. The notebook explicitly notes that model probability represents model confidence rather than certainty about the customer's true segment.

Model Card

Item	Description
Model	CategoricalNB
Task	Four-class customer segment prediction
Dataset	Train.csv supplied to the notebook
Predictors	9 non-ID features
Primary selection metric	5-fold CV Macro F1
CV Macro F1	0.4845
Locked-test Macro F1	0.4853
Intended use	Educational and analytical segmentation support
Out-of-scope	Unsupervised clustering, autonomous high-stakes decisions, or treating probabilities as certainty

Responsible Use, Limitations and Reproducibility

The notebook includes privacy and fairness considerations and removes the direct customer identifier from predictors. However, the dataset documentation available inside the notebook does not fully establish the provenance of Spending_Score and Var_1; these fields remain flagged for confirmation. No explicitly documented psychographic variables were available, so conclusions about psychographic effects cannot be drawn.

The final model has several important limitations.

1. Moderate Predictive Performance

The locked-test Macro F1 is: 0.4853

and accuracy is: 51.18%

This means that the model does not provide highly reliable classification for all customers.

2. Uneven Class Performance

Performance differs substantially among classes.

Segment D has an F1 of **0.6840**, whereas Segment B has an F1 of only **0.2785**.

Therefore, the aggregate Macro F1 hides significant differences between segments.

3. Weak Segment B Recall

Segment B recall is only: 0.2231

This indicates that many actual B customers are classified as another segment.

The B → C confusion pattern, with **157 cases**, is particularly significant.

4. Feature Provenance

The notebook does not fully establish the provenance of Spending_Score and Var_1. This must be resolved before any practical deployment.

5. No Psychographic Variables

The dataset does not contain explicitly documented psychographic-only features. Consequently, the study cannot evaluate whether psychographic information would improve segmentation.

6. Probability Is Not Certainty

Although higher confidence generally corresponds to higher observed accuracy, the relationship is not perfectly monotonic.

Therefore:

$P(\text{class}) \neq \text{guaranteed correctness}$

The probability should be interpreted as the model's estimated confidence rather than a certainty statement.

The modest locked-test Macro F1 of 0.4853 and the uneven class-wise performance demonstrate that the model should be treated as an educational analytical prototype. In particular, the weak recall for Segment B indicates a material risk of systematically missing that class. External validation, stronger provenance documentation, subgroup analysis and potentially richer features would be needed before operational deployment.

Reproducibility and Submission Evidence

The notebook fixes the random seed at 42, uses an 80:20 stratified split, applies five-fold StratifiedKFold on the training partition, keeps preprocessing within pipelines, excludes customer IDs from predictors and evaluates the frozen model once on the locked test set. It also saves artifacts including the split manifest, feature manifest, model configuration, final model-selection summary, test results, confusion matrix, error analysis, confidence summary, validation coverage-error analysis, test review analysis and new-customer prediction.

The final submission checklist initially reported three incomplete items—confidence analysis, validation review analysis and test review analysis. The later 'STEP 13 — COMPLETE CONFIDENCE & REVIEW ANALYSIS' cell successfully generated and saved all three required artifacts. Thus, the confidence/review portion of the workflow was completed after the initial checklist check.

Discussion

The experimental results provide several important observations about the customer-segmentation problem.

First, the comparison with the DummyClassifier establishes that the Naive Bayes models learn meaningful predictive patterns from the customer attributes. The DummyClassifier achieved only **0.1097 Macro F1**, whereas the selected CategoricalNB achieved **0.4845** in cross-validation.

Second, the representation of the features has a significant effect on model performance. The initial CategoricalNB representation was invalid because the transformed values did not satisfy the model's non-negative categorical requirement. After correcting the representation, the model produced valid and reproducible cross-validation results.

Third, CategoricalNB and BernoulliNB perform very similarly. Their Macro F1 scores are 0.4845 and 0.4801 respectively. The difference of approximately 0.0045 is relatively small. Therefore, the selection of CategoricalNB should be understood as the result of applying the predefined selection criterion rather than as evidence that CategoricalNB is universally superior.

Fourth, the feature-group experiment indicates that demographic features are highly informative for this dataset. The demographic-only configuration achieved a Macro F1 of **0.4802**, very close to the combined model's **0.4845**. Behavioral features alone, however, achieved only **0.2726 Macro F1**.

Fifth, class-wise analysis reveals that overall performance is not evenly distributed. Segment D is predicted substantially better than Segment B. The model's overall Macro F1 therefore needs to be interpreted together with the individual class metrics.

Finally, the confidence analysis demonstrates that the model can be used with an abstention mechanism. At the selected threshold of 0.55, the system accepts only sufficiently confident predictions and sends uncertain cases for review. This provides a more cautious operational strategy than automatically accepting every model prediction.

Conclusion

This study develops a complete supervised customer-segmentation prediction workflow using Naive Bayes classifiers and a leakage-aware evaluation design.

The dataset contains **8,068 customer records**, nine predictors and four target segments: A, B, C and D. After auditing the data, handling missing values and excluding the customer identifier from the predictor matrix, four models were compared using identical five-fold stratified cross-validation.

Among the evaluated models, **CategoricalNB achieved the highest mean Macro F1 of 0.4845**, followed closely by BernoulliNB at 0.4801. The combined nine-feature representation also achieved the best feature-group performance.

The model achieved the following locked-test results:

- **Accuracy:** 51.18%
- **Macro F1:** 0.4853
- **Weighted F1:** 0.4955

The close agreement between cross-validation Macro F1 (**0.4845**) and locked-test Macro F1 (**0.4853**) indicates that the aggregate evaluation was reasonably consistent between validation and unseen data.

However, the class-wise results reveal substantial variation. Segment D achieved the strongest F1 of **0.6840**, while Segment B achieved only **0.2785**. The low recall of Segment B and the large B → C confusion pattern demonstrate that the model has difficulty separating certain customer groups.

The confidence analysis provides an additional mechanism for managing this uncertainty. A validation-derived threshold of **0.55** resulted in 55.38% validation coverage and 61.89% accuracy among accepted predictions. When applied to the locked test set, the threshold resulted in **910 automatic predictions and 704 review cases**, corresponding to **56.38% automatic coverage and 43.62% review coverage**.

Overall, the study demonstrates a reproducible workflow that combines **data auditing, leakage prevention, model comparison, feature-group analysis, locked-test evaluation, class-wise analysis, confidence assessment and human review**.

The final model should nevertheless be regarded as an **educational and analytical prototype rather than a production-ready autonomous decision system**. Before real-world deployment, additional work would be required, particularly around feature provenance, external validation, calibration, subgroup evaluation, improvement of Segment B performance and the inclusion of richer customer information.

The most appropriate interpretation of the current results is therefore that the model can serve as a **segmentation-support tool**, where predictions—especially low-confidence predictions—are treated as analytical evidence that can assist further review rather than as unquestionable customer classifications.

Github :

https://github.com/Shobana0212/Lab4_Probabilistic-Customer-Segmentation-and-Segment-Prediction.git